

SUBSTRATES

The chemicals involved in the reaction may initially be drawn to separate areas of the enzyme surface by weak attractive forces.

ENZYME

As the substrates bond to the enzyme, chemical changes allow them to overcome the energy barrier that prevents a reaction, forming a product that is then released.

Catalytic process

Catalysis involves changing the structure of two or more chemicals to allow a reaction between them. In some cases, the barrier to reaction appears to be overcome only through quantum tunneling.

TUNNELING FOR A REACTION

Chemicals called enzymes are found throughout our bodies, aiding processes such as digestion (the breaking down of food into useful nutrients). They are thought to work by lowering the energy barrier between molecules called substrates so that they undergo reactions, but the precise way in which they do this remains poorly understood. One theory is that they do this by creating conditions in which electrons are able to bridge the gap between molecules using quantum tunneling (see pp.70–71).

